

Data Analyst Job Postings Analysis

Salary Estimates, Experience Requirements, and Programming Language Categories

Dataset sample size: 400 job postings

Brief Introduction

This project analyzes a sample of Data Analyst job postings to explore how salary estimates relate to minimum years of experience and programming language requirements. The goal is to identify simple patterns in the data while recognizing that the sample is limited, and the findings are descriptive rather than conclusive.

Data Preparation Notes

The minimum experience field was cleaned so that numeric values could be plotted; entries such as NA, blanks, or non-numeric text were treated as missing. The Data Languages field was standardized into four categories: R, Python, Both, and Neither.

Measure	Value
Total postings in sample	400
Rows with usable numeric experience values	312
Programming language categories	R, Python, Both, Neither

Years of Experience Analysis

Figure 1 presents a scatter plot of Minimum Years Experience on the x-axis and Median Salary Estimate on the y-axis.

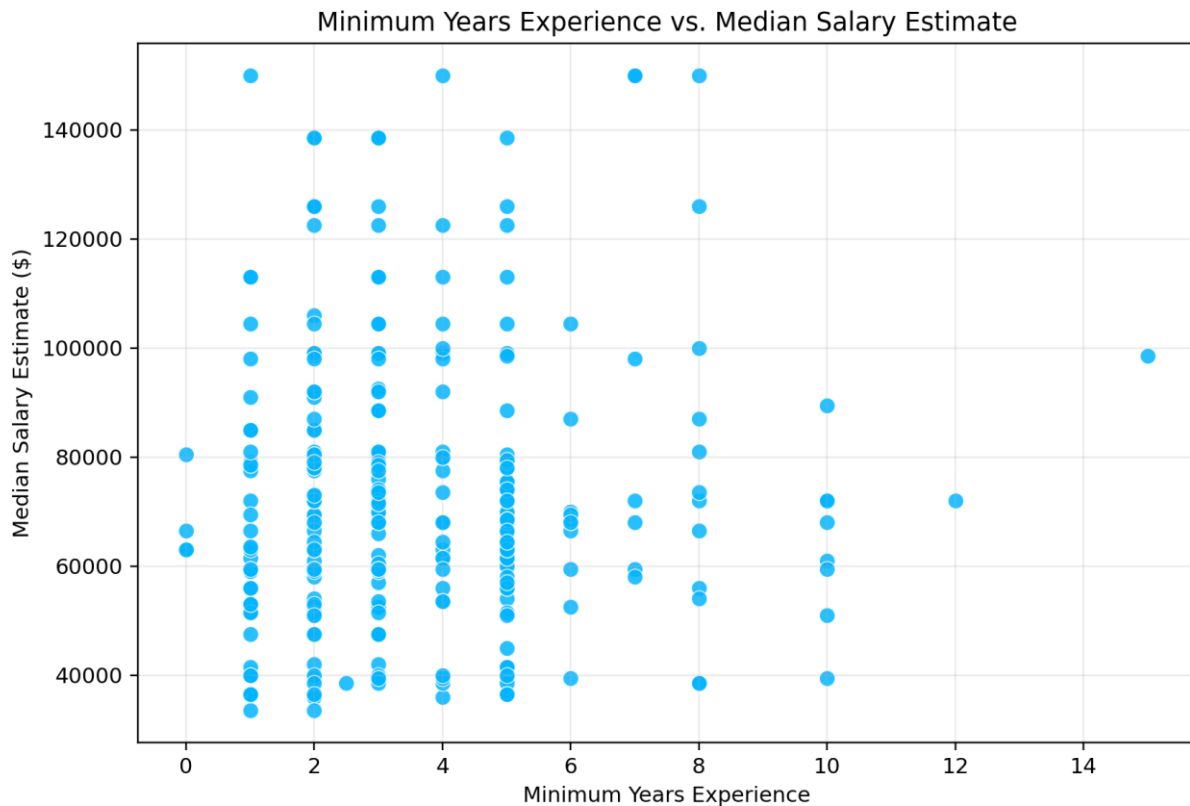


Figure 1. Minimum Years Experience vs. Median Salary Estimate

Interpretation:

The scatter plot shows a general tendency for salary estimates to vary across different experience levels, but the relationship does not appear to be strongly linear. Some job postings with relatively low experience requirements still offer high salaries, while some postings requiring more experience have only moderate salary estimates. This suggests that salary is likely influenced by additional factors such as job title, industry, location, and company type. Since the dataset is only a sample and only 312 postings had usable numeric experience values, the plot should be interpreted cautiously.

Programming Language Analysis

Figure 2 presents a box plot of Median Salary Estimate grouped by standardized Data Languages categories.

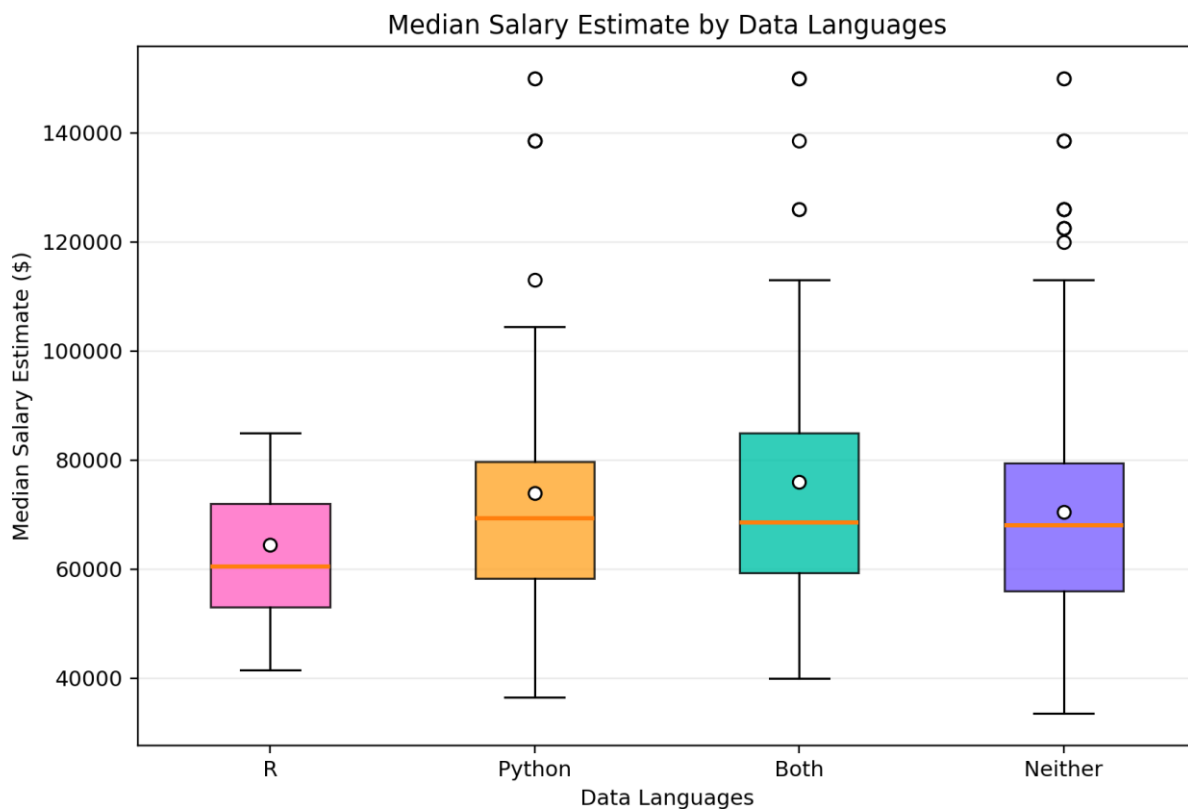


Figure 2. Median Salary Estimate by Data Languages

Interpretation:

The box plot shows that median salary estimates are relatively similar across the Python, Both, and Neither categories, with substantial overlap in salary ranges. The R-only category appears somewhat different in this sample, but it includes far fewer postings than the other groups, making that comparison less reliable. Overall, the plot suggests that programming language category alone does not clearly separate salary outcomes in this dataset. The results should be viewed as descriptive because the sample is modest in size and the category counts are uneven.

Conclusion

In summary, the visual analysis identifies broad variation in salary estimates across experience requirements and programming language categories, but it does not show a strong single-factor explanation for salary differences. The findings support the idea that salary estimates in Data Analyst postings are likely shaped by multiple job characteristics, and future analysis with a larger dataset could include additional variables such as location, industry, company size, and job seniority.

Workshop 3 Reflection

The workshop presented some initial challenges; however, with continued practice, the processes became increasingly manageable. I developed a solid understanding of how to draft effective prompts and define clear instructions for AI tools. The data analysis capabilities demonstrated through ChatGPT were particularly impressive, as was the ability to generate a comprehensive report within minutes, this stood out as one of the most remarkable aspects of the experience.

The concept of API keys was new to me, and I recognize that one of the key challenges going forward will be acquiring and practicing with these tools using real-world data. This will require dedicated time, hands-on experience, and investment in paid versions of the relevant platforms. Nevertheless, despite the initial complexity of the workshop, it was deeply rewarding to see a polished, structured report as the final output.

While only a limited number of indicators were used during this session, it is clear that with high-quality and relevant data, the scope can be significantly expanded to incorporate a much wider range of indicators, further enhancing the depth and value of the analysis.